

SHOURYA THAPLIYAL

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LinkedIn: [linkedin.com/in/shourya-thapliyal-53630b153](https://www.linkedin.com/in/shourya-thapliyal-53630b153) | Portfolio: shouryat32.github.io/Portfolio

Visa Status: 485 Graduate Work Visa (Valid until January 2029)

PROFESSIONAL SUMMARY

Data Scientist with 2 years software engineering experience at Tata Consultancy Services delivering banking solutions for Commonwealth Bank of Australia. Master of Data Science graduate (2025) from University of Melbourne specializing in machine learning, natural language processing, and AI agents. Achieved 84% ML model accuracy and 95% test automation coverage in recent projects. Integrated 15+ REST APIs in production environments. Published researcher in machine learning for cardiac arrhythmia detection with expertise in financial services and healthcare analytics.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Java, JavaScript, HTML, CSS, Node.js

Machine Learning & AI: scikit-learn, XGBoost, Random Forest, TensorFlow, PyTorch, Keras, Hugging Face Transformers, Neural Networks, Ensemble Methods, Statistical Modeling, Hyperparameter Tuning, Cross-Validation

Natural Language Processing: NLTK, spaCy, TF-IDF, Word Embeddings, Sentiment Analysis, Text Classification, NLP Pipelines

Data Engineering & Processing: Pandas, NumPy, ETL Pipelines, Data Modeling, Data Cleaning, Feature Engineering, Data Transformation, Qualtrics API, REST APIs

Data Visualization & BI: Power BI, Tableau, Plotly, Matplotlib, Seaborn, Interactive Dashboards, Data Storytelling, KPI Tracking

Databases: PostgreSQL, MySQL, SQL Server, CouchDB, NoSQL, Database Design, Query Optimization, Data Warehousing

Cloud & DevOps: Docker, Ansible, NGINX, CI/CD, Cloud Computing, Infrastructure as Code, Containerization, Deployment Automation

Development Tools: Git, GitHub, JIRA, Selenium, Jupyter Notebooks, VS Code, Agile/Scrum, Test Automation, TDD, Version Control

PROFESSIONAL EXPERIENCE

Software Engineer

Tata Consultancy Services | Commonwealth Bank of Australia

Bangalore, India | January 2021 – December 2022

- Engineered 15+ REST API integrations for payment processing and account management, reducing integration time 30% through reusable frameworks
- Achieved 95% test coverage and reduced regression testing time 60% using Selenium and Python automation
- Built responsive JavaScript components for NETBANK Preference Centre serving 100,000+ users
- Collaborated in Agile/Scrum teams delivering features within sprint timelines

KEY PROJECTS

Phoenix Australia Mental Health Analytics Pipeline

University of Melbourne | March 2025 – December 2025

- Designed and deployed automated data pipeline integrating Qualtrics API to collect and analyze mental health course feedback from 500+ participants across 12 cohorts, reducing manual data processing time by 85%
- Developed interactive Power BI dashboards with Python visualizations (Plotly, Matplotlib) tracking 15+ key metrics including participant outcomes, course completion rates, engagement levels, and longitudinal wellbeing improvements for stakeholder reporting
- Implemented robust ETL workflows using Python (Pandas, NumPy) and SQL to clean, transform, and validate participant survey data, ensuring data quality and integrity for academic research and program evaluation
- Performed statistical analysis and data modeling to identify trends in mental health outcomes, providing actionable insights that informed program improvements and evidence-based treatment approaches
- Collaborated with clinical researchers and program coordinators to define KPIs, translate research questions into analytical frameworks, and present findings to multidisciplinary stakeholders

Machine-Generated Text Detection (Kaggle Competition)

University of Melbourne | January 2024 – March 2024

- Achieved 84% accuracy using ensemble methods (Random Forest, XGBoost, Neural Networks), ranking top 10%
- Implemented feature engineering with TF-IDF vectorization and word embeddings

Social Media Analytics Platform

University of Melbourne | August 2024 – November 2024

- Built cloud analytics system using Docker, CouchDB, NGINX processing 100,000+ monthly posts
- Implemented sentiment analysis correlated with Australian crime data using NLP techniques

Cardiac Arrhythmia Prediction System

Nitte Meenakshi Institute | 2019 – 2020

- Developed ML classification model achieving 72% accuracy using Python and PhysioNet dataset
- Published research at international conference (Springer ERCICA Volume 1)

EDUCATION

Master of Data Science

University of Melbourne | Melbourne, Australia | February 2023 – February 2025

Coursework: Machine Learning, Deep Learning, Big Data Analytics, Statistical Modeling, Cloud Computing, NLP

Bachelor of Engineering (Computer Science)

Nitte Meenakshi Institute of Technology | Bangalore, India | 2016 – 2020

Specialization: Software Development, Data Structures, Algorithms, Database Management

CERTIFICATIONS

- Hugging Face: AI Agents Fundamentals, Model Context Protocol, Automation in Production (2025)
- Google Data Analytics Professional Certificate (2025)
- Intermediate R Programming – DataCamp (2023)
- SQL Basic and Intermediate – HackerRank (2025)

PUBLICATIONS

Thapliyal, S., et al. (2020). "Cardiac Arrhythmia Prediction Using Machine Learning." Springer ERCICA Volume 1.